

Problem 1

- a) The skew of -0.67 rules out the normal and student's t distribution, as those typically produce a sample skew close to 0. It rules out the lognormal distribution as that would be expected to produce a positive skew. This only leaves the following families of distributions as plausible: NIG, skewed t, and generalized hyperbolic. The sample excess kurtosis of 2.36 is plausible for these families.
- b) There are 26 observations that fall below the Normal's 1% quantile. If the distribution were truly normal, we would expect 10 observations to fall below the 1% quantile given the sample size of 1000.
- c) As can be seen in the histogram and the QQ-plot (figures/problem1_fit.png), the fitted Normal would severely understate the probability of large negative outcomes. The forced symmetry of the fitted Normal has no way of picking up that the left tail of the distribution is longer than the right tail (skew = -0.67). The fitted normal would also understate both extreme negative and positive outcomes, as it assumes an excess kurtosis of 0 (vs actual of 2.36). The QQ plot shows that the understatement of probability is larger on the left tail side compared to the right tail side due to the left skew.

Problem 2

- a) The scatter shows a positive linear relationship between x and y. While the typical assumption until proven otherwise is that errors are normally distributed, the title of this problem, "A Regression Whose Errors Are Not Normal", suggests that the errors are not normally distributed. This is consistent with the observation that the scatter shows a tight core of data points with occasional large deviations (e.g. around $0 < x < 1$). This suggests that the tails of the error distribution might be larger than those of a Normal distribution, which hints at the errors potentially following a student's t distribution.
- b) This data violates the OLS assumption that errors are normally distributed. While the slope estimate may still be accurate, the breaking of this assumption can invalidate potential hypothesis testing and confidence intervals. The violation will affect the standard error of the slope by inflating it and making it less trustworthy. Heavier tails cause more extreme outliers. These inflate the standard error as residuals are squared. The standard error then reflects the outliers disproportionately to the underlying population variance.
- c) I predicted that the slope "may still be accurate", which is the case here (betas of 2.99 vs 3.02). OLS and MLE under a Normal error produce identical alphas and

betas (2.99), while the MLE under a Student's t error produces slightly different parameters (beta of 3.02). The violation did not move the slope a lot which is in line with my prediction that the slope may still be accurate. The student's t distribution is, like the normal distribution, symmetric around zero. The change in assumption affects how much weight large residuals get in the fitting process. This leads to the slopes approximating each other. Given enough samples, it is likely to observe similar parameter estimations. At smaller sample sizes, the heavier tails of the student's t distribution could have a larger effect on the slope estimation.

- d) The rejected model is wrong about the tail behavior of the error distribution. It understates the probability of extreme errors.
- e) The normal is wider at 95% while the student's t is wider at 99.5%. The student's t distribution's scale is lower than the standard deviation of the normal, leading the t to predict a tighter spread around the center. The t's low nu of 3.63 means the tails decay slower than the ones of the Normal. One would prefer the student's t error for setting a capital buffer, as a capital buffer is typically more concerned with the 99.5% level rather than the 95% level, and the Normal understates extreme errors.

Problem 3

- a) I expect the Pearson and Spearman coefficients to disagree most on (x1, x2) as the relationship is non-linear, but monotonic. I expect them to agree the most on the pairs (x1, x4) and (x3, x4) as there seems to be no correlation between the pairs, meaning the two would agree on a 0-ish correlation.
- b) The largest gap is in pair (x1, x2) at 0.17.
- c) As predicted in a), the gap is due to the fact that the relationship is non-linear but monotonic. Spearman is the honest description as it captures the non-linearity of the relationship better. Pearson is answering how strong the linear association between the variables is.

Problem 4

- a)
$$\text{Var}(x_2 | x_1) = \Sigma_{(x_2, x_2)} - \Sigma_{(x_2, x_1)} \cdot \Sigma_{(x_1, x_1)}^{-1} \cdot \Sigma_{(x_1, x_2)}$$

block 1 = x2 & block 2 = x1

$$\text{factor} = (\Sigma_{(x_2, x_2)} - \Sigma_{(x_2, x_1)} \cdot \Sigma_{(x_1, x_1)}^{-1} \cdot \Sigma_{(x_1, x_2)}) / \Sigma_{(x_2, x_2)}$$

factor: $1 - \rho^2$

factor for this data: 0.3622
- b) No, the factor does not depend on the value of x1 as every expression in the block is built from the covariance matrix and does not rely upon any observed values. It is easier to see in the simplified formula, the only variable is rho, there is no observed

values in the formula. This is the block equivalent to rho from the equation above:

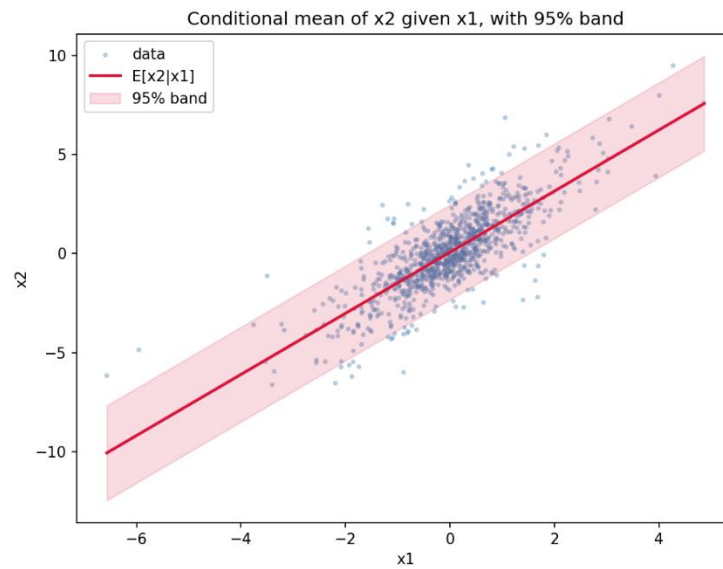
$$\Sigma_{(x2,x1)} \cdot \Sigma_{(x1,x1)}^{-1} \cdot \Sigma_{(x1,x2)}$$

c) $E(x2 | x1) = \mu(x2) + \Sigma_{(x2,x1)} \cdot \Sigma_{(x1,x1)}^{-1} \cdot (x1 - \mu(x1))$

coefficient on x1: $\Sigma_{(x2,x1)} \cdot \Sigma_{(x1,x1)}^{-1}$

In regression terms it is the slope obtained by regressing x2 on x1.

Plot:



d) fraction inside band = 0.9350 (935 / 1000)

e) Coverage by distance of x1 from its mean:

Bucket	N	Coverage
Within 1 sd	759	0.9565
1 tp 2 sd	190	0.8579
Beyond 2 sd	51	0.9020

f) The non-flat coverage means that the conditional variance of x2 given x1 changes across the range of x1. This breaks the assumption that under the bivariate Normal, the variance of x2 given x1 does not depend on x1. The conditional mean does not depend on constant conditional variance and is still a reasonable description of x2's average, surviving the breaking of the assumption mentioned above. The band fails as a constant width applied over the entire distribution only makes sense if the conditional variance were constant across x1.

Problem 5

a) The PACF cuts off at the second lag, the ACF decays. This implies an AR(2) process.

b) The rule used was “PACF cutting off while ACF decays indicates an AR process and vice versa. The cutoff lag determines the order.” The significance band used was the $\pm 1.96/\sqrt{n}$

c) Model Fits:

Model	AICc
AR(1)	1418.5870
AR(2)	1372.4383
AR(3)	1374.3432
MA(1)	1389.0843
MA(2)	1381.2316
MA(3)	1374.1165

d) The AICc selects the AR(2) model. It did match my prediction. The AICc is lowest for the AR(2) model.

e) AICc penalizes each added parameter and therefore favors smaller models. It trades off model complexity vs in-sample accuracy/fit. R^2 does not have such a penalty. Adding parameters artificially inflates R^2 as more parameters fundamentally/mathematically allow for a better in-sample fit. The penalty is useful to avoid overfitting to the available sample.